

Internal anatomy of brainstem II: major tracts and cranial nerve nuclei

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Make.
Heal.
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Session Objectives

- 1) Understand the general route of the corticospinal tract, dorsal-column medial lemniscus, antero-lateral tract, and cerebellar peduncles.
- 2) Identify from external landmarks and horizontal section the placement of corticospinal tract, dorsal-column medial lemniscus, antero-lateral tract, and cerebellar peduncles.
- 3) Understand the position of the cranial nerve nuclei with respect to their general functions.

Major tracks in the brainstem

Descending from the cerebral cortex

corticospinal to spinal cord (most decussate at caudal medulla)
for voluntary movement

Ascending to the cerebral cortex

dorsal column medial lemniscus: touch, proprioception for limbs

DC → internal arcuate fibers (decussate in medulla) → ML

spinothalamic: pain, temperature for limbs (decussate in spinal cord)

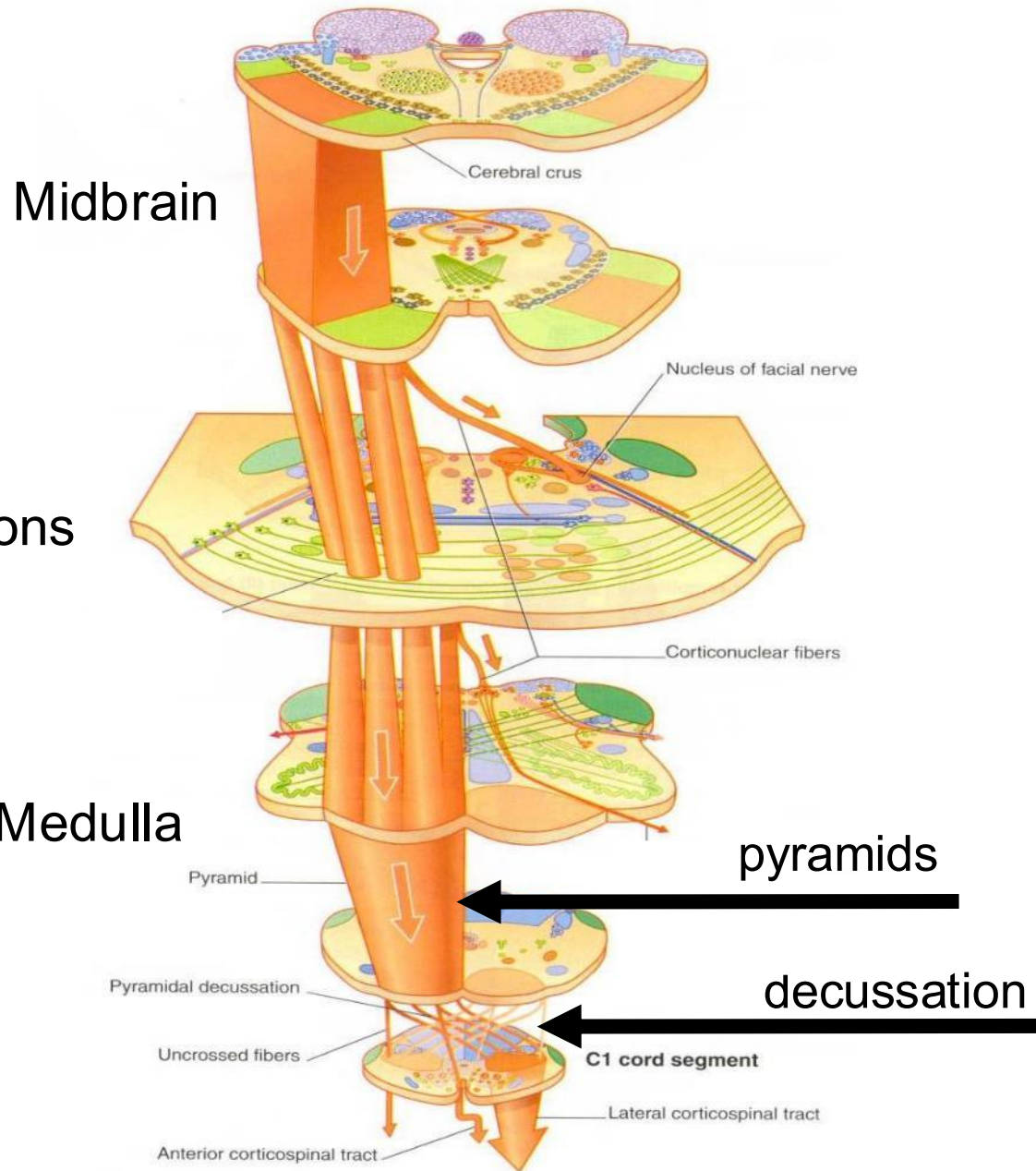
Cerebellar peduncles (motor coordination)

inferior peduncle: afferent to cerebellum (at medulla)

middle peduncle : afferent to cerebellum (at pons)

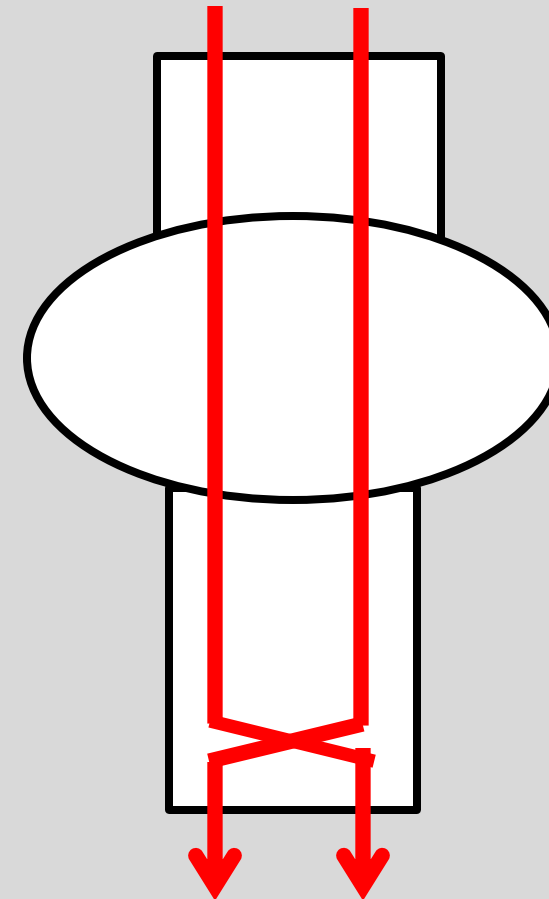
superior peduncle: efferent of cerebellum (at midbrain)

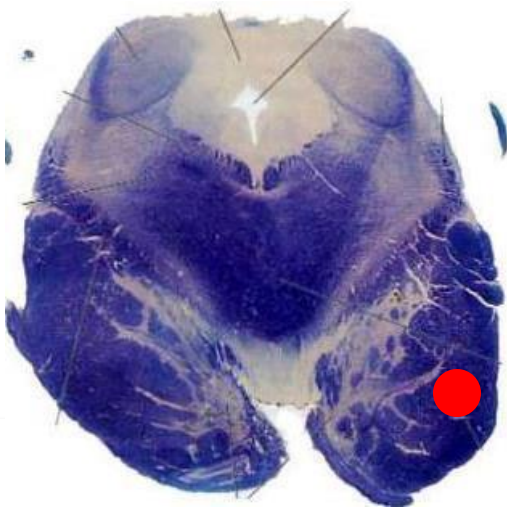
Corticospinal tract



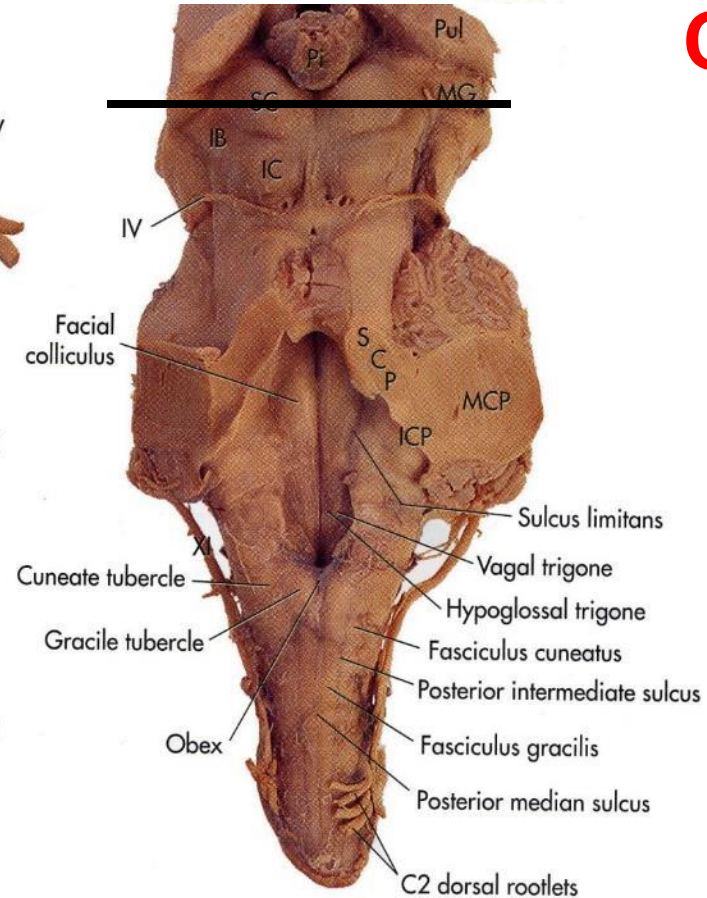
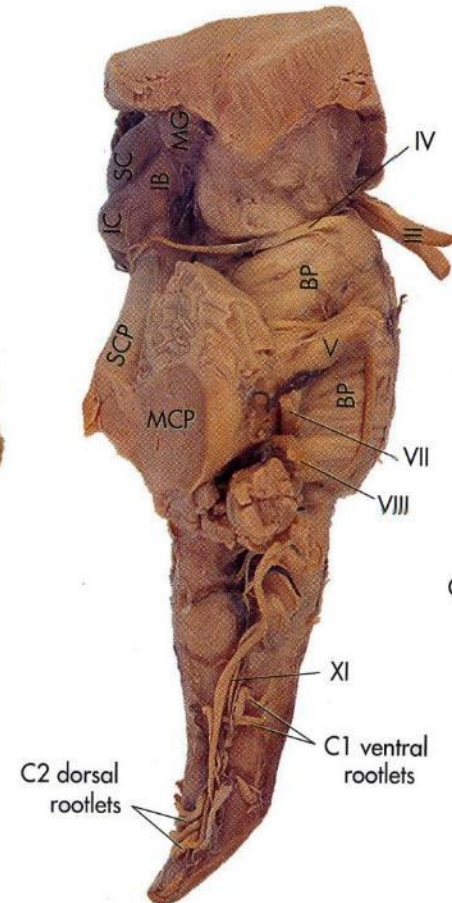
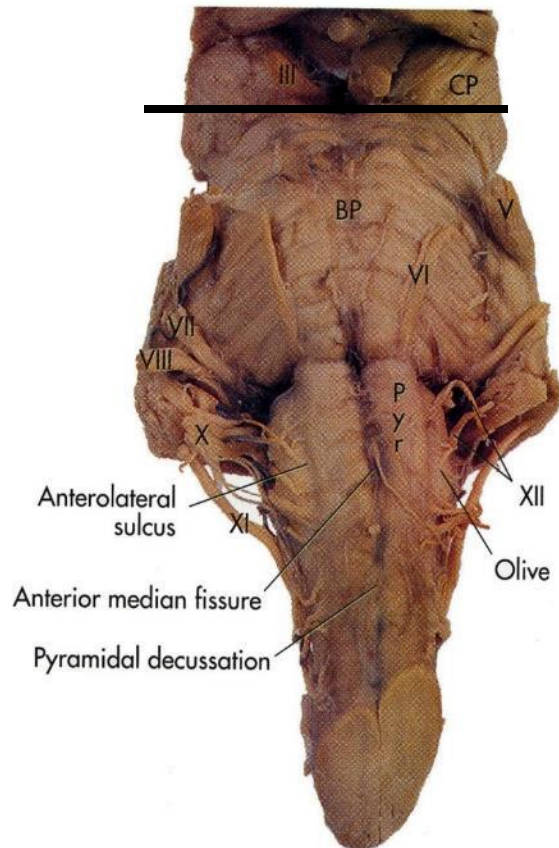
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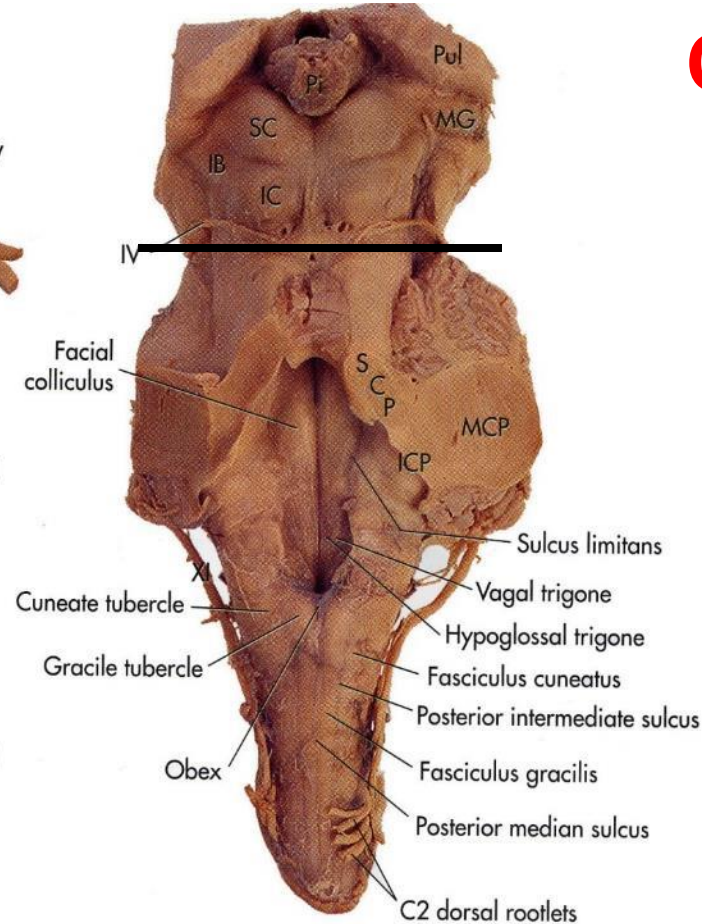
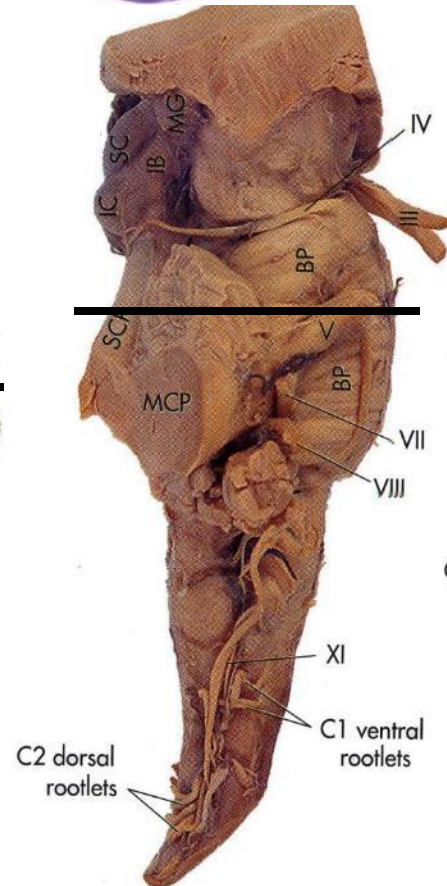
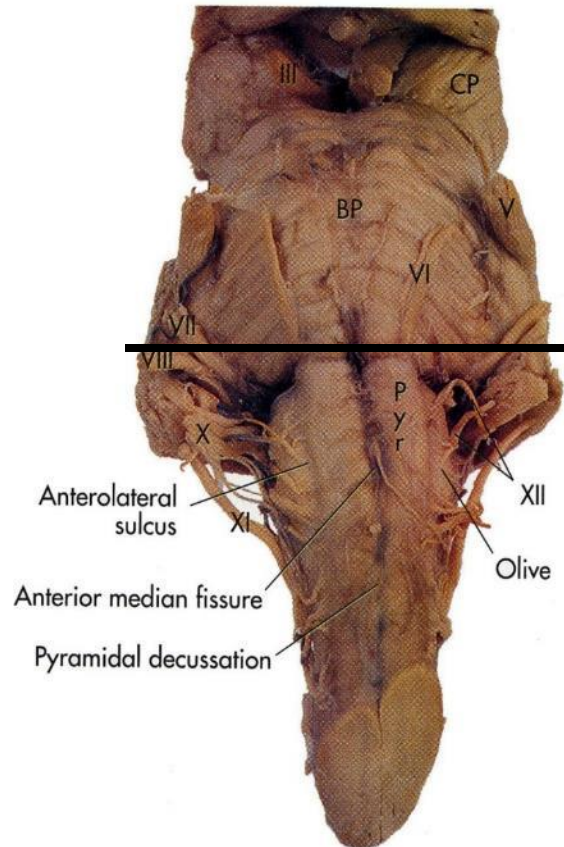


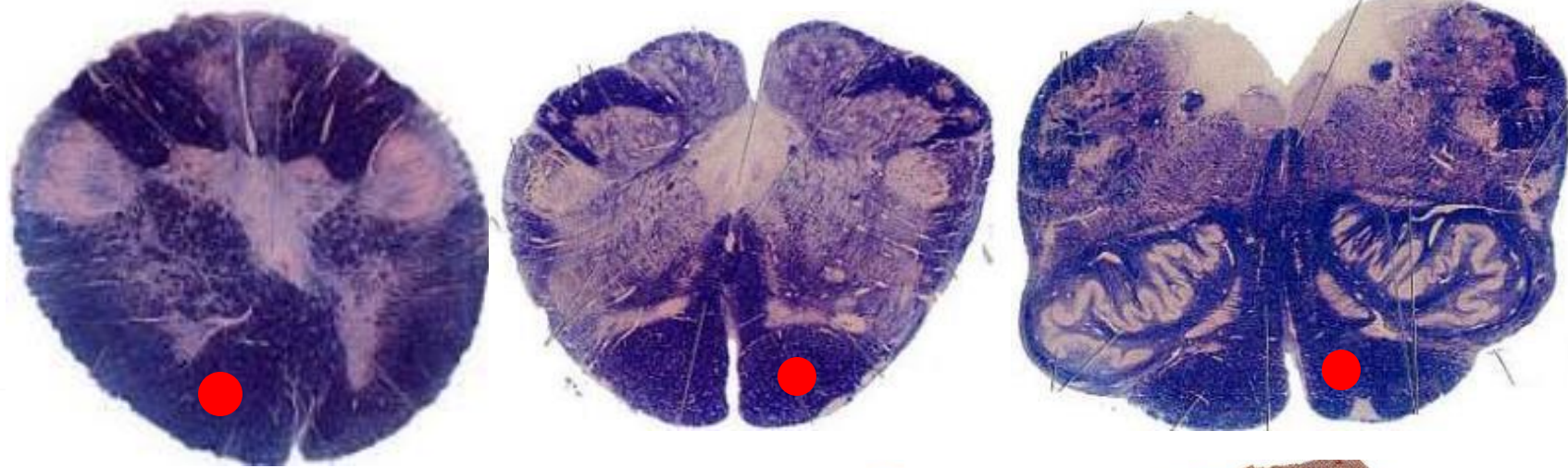


Corticospinal tract



Corticospinal tract

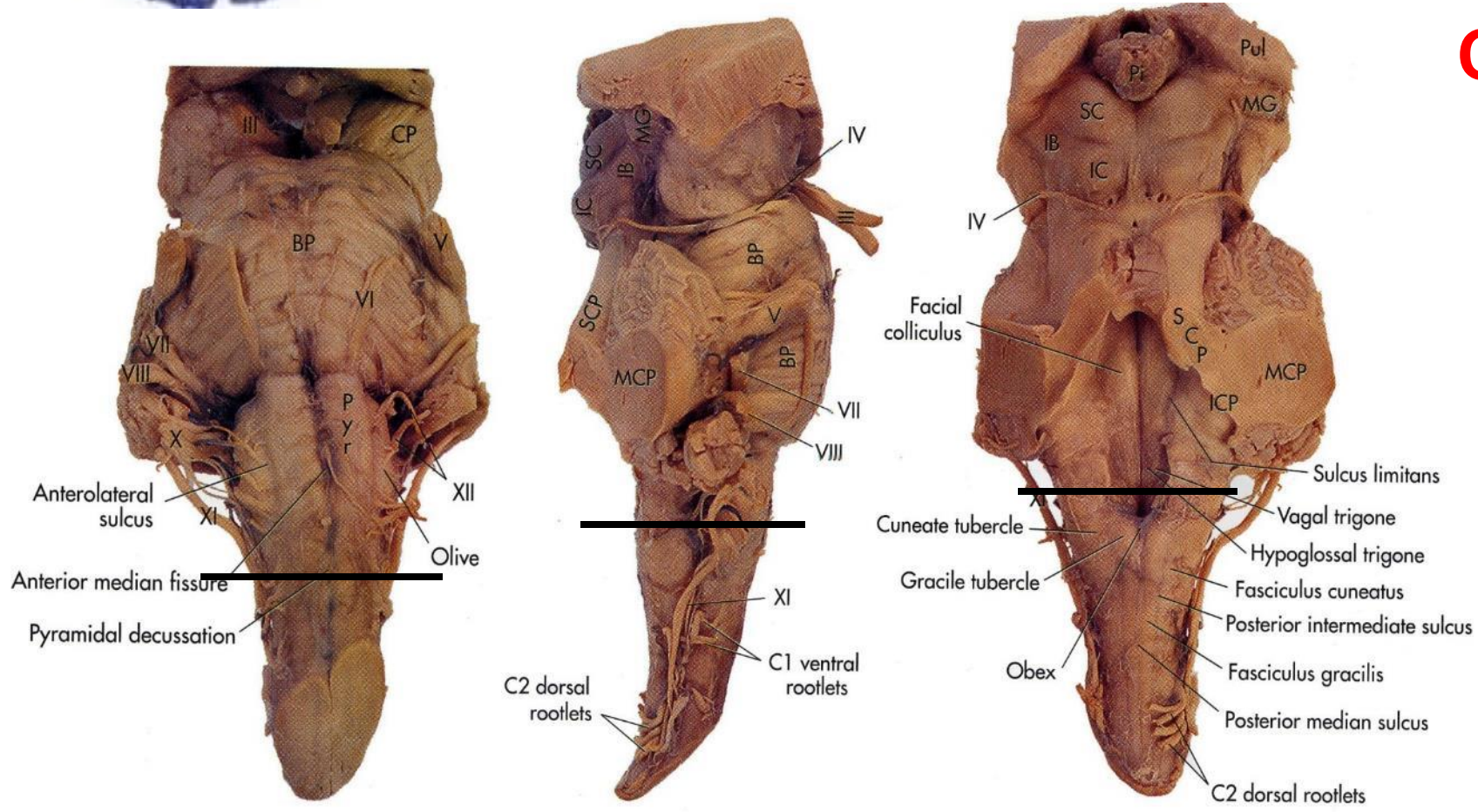




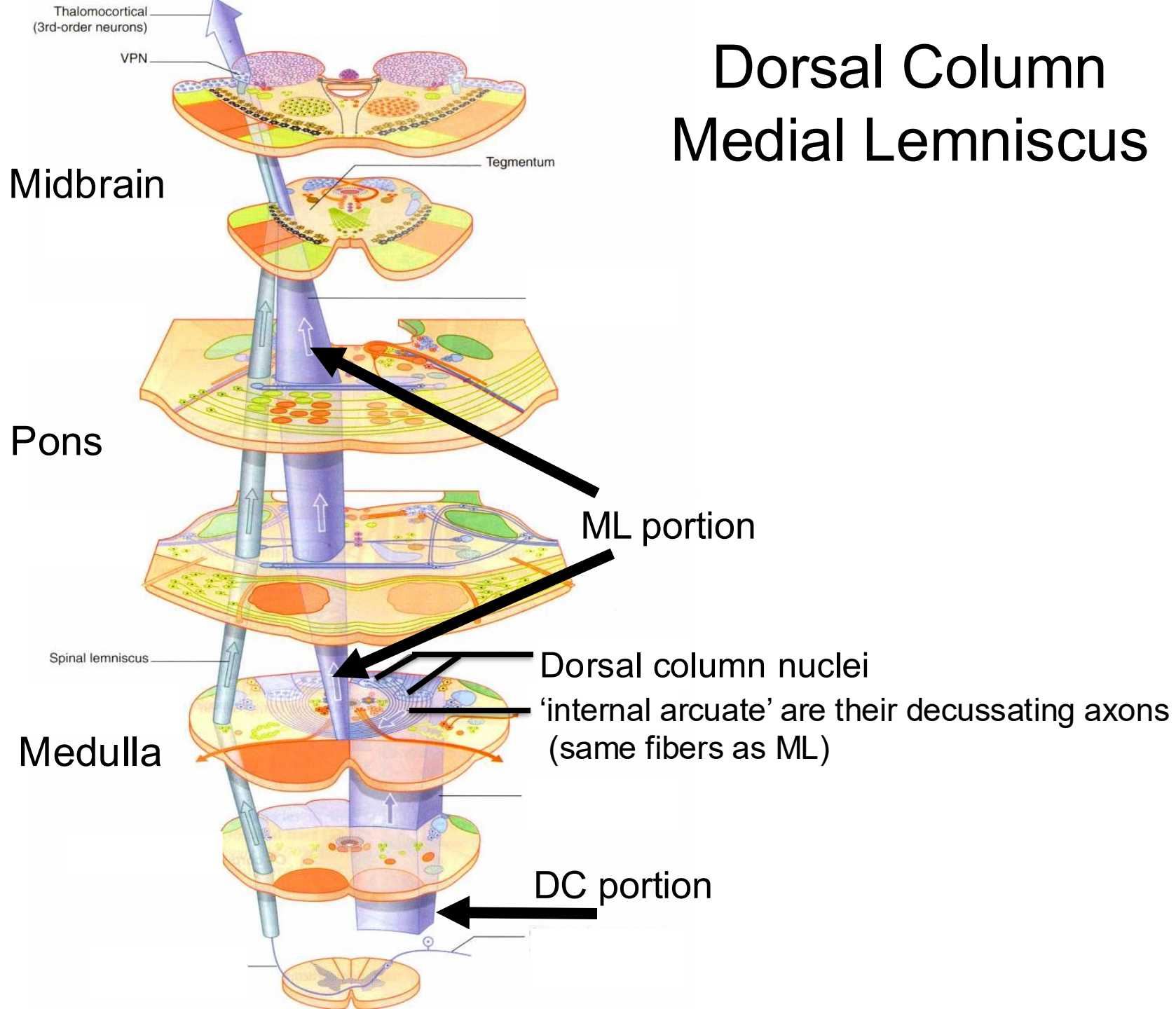
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Corticospinal tract

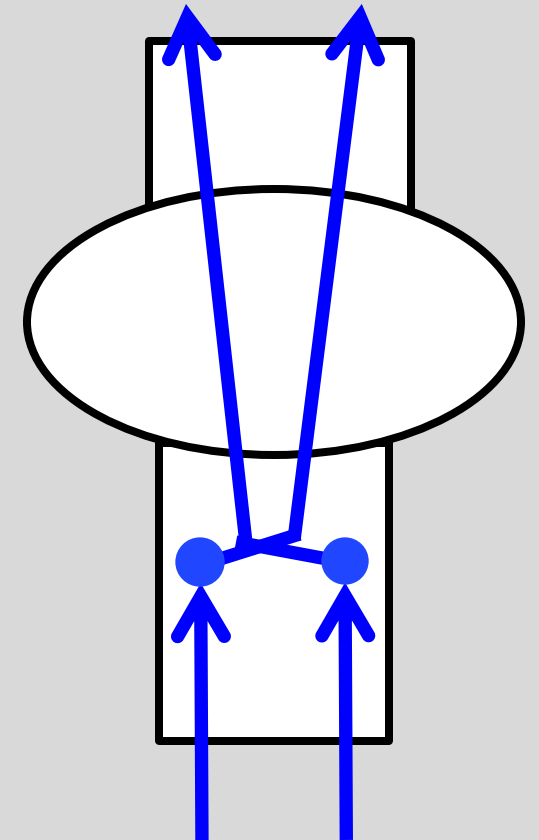


Dorsal Column Medial Lemniscus



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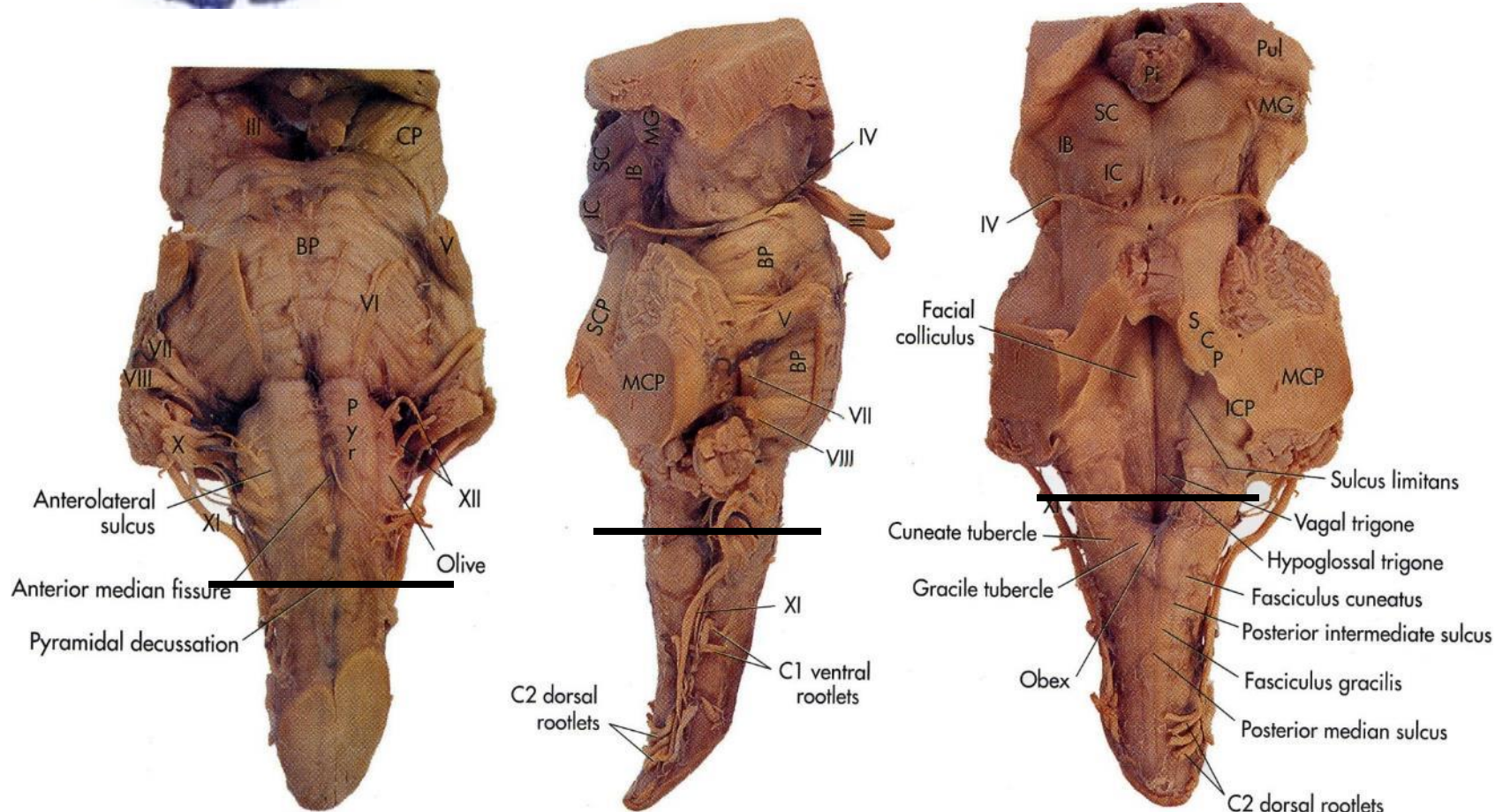
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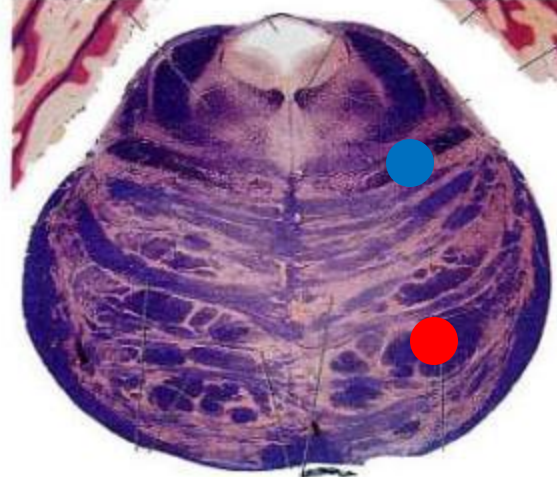
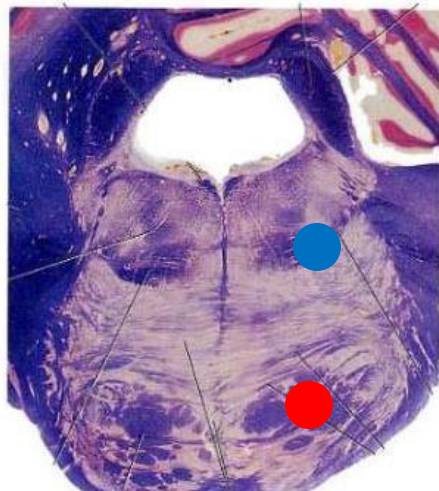
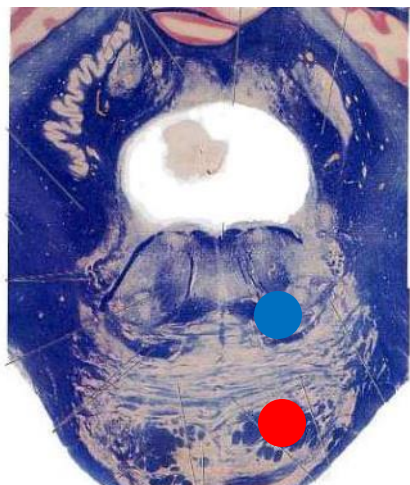


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corticospinal
DCML

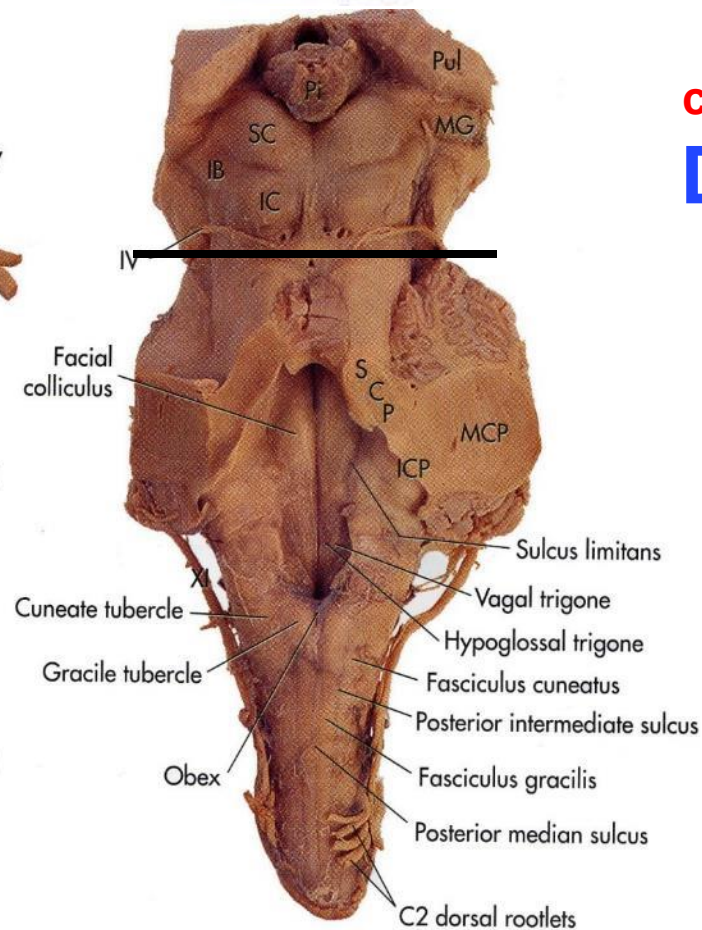
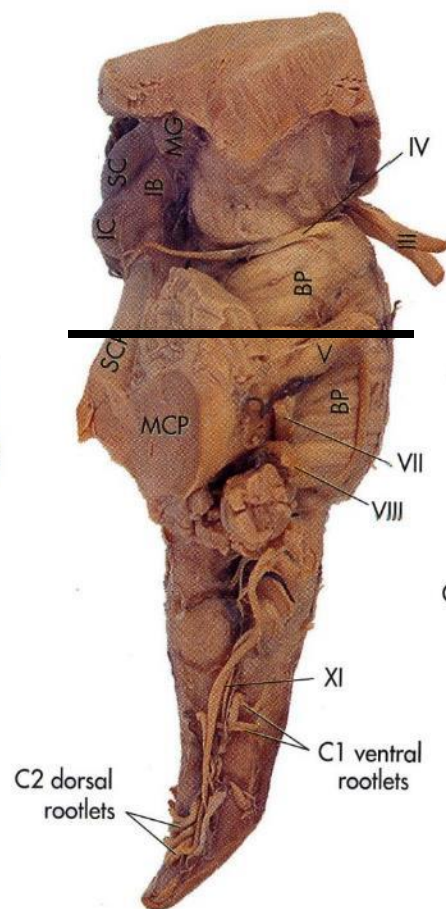
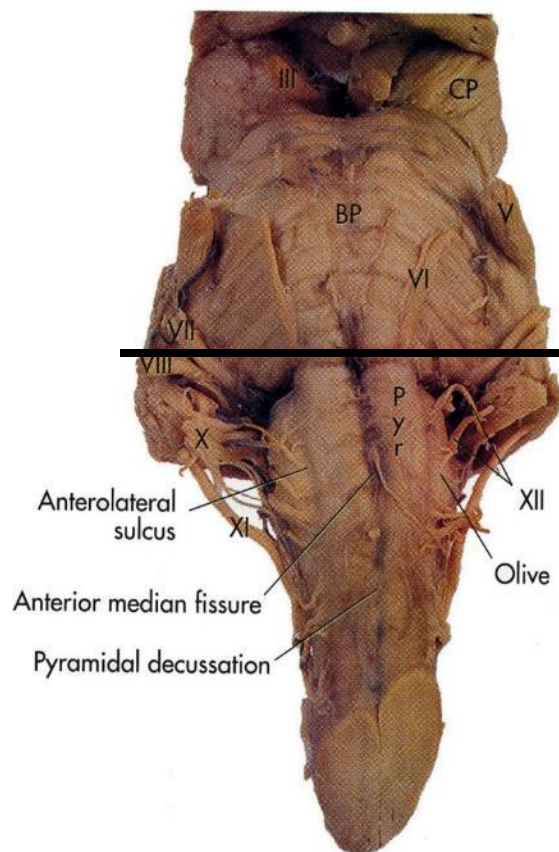


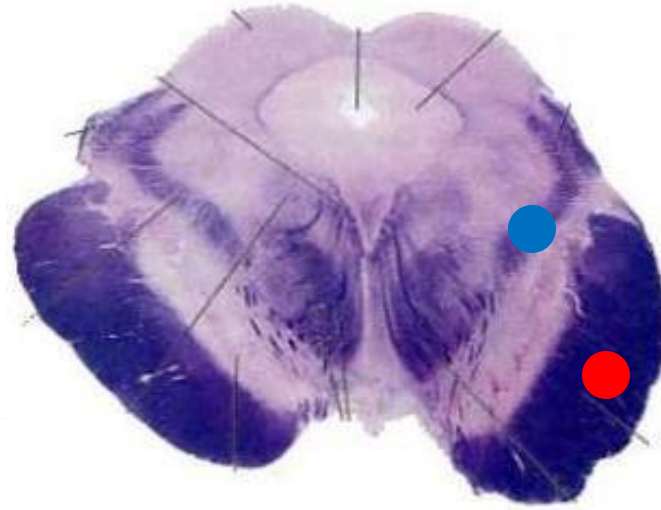
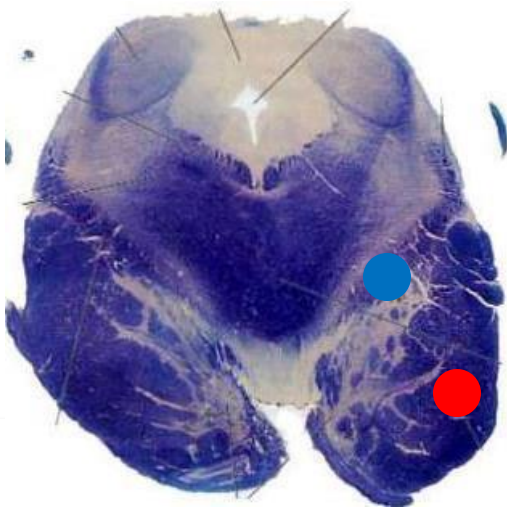


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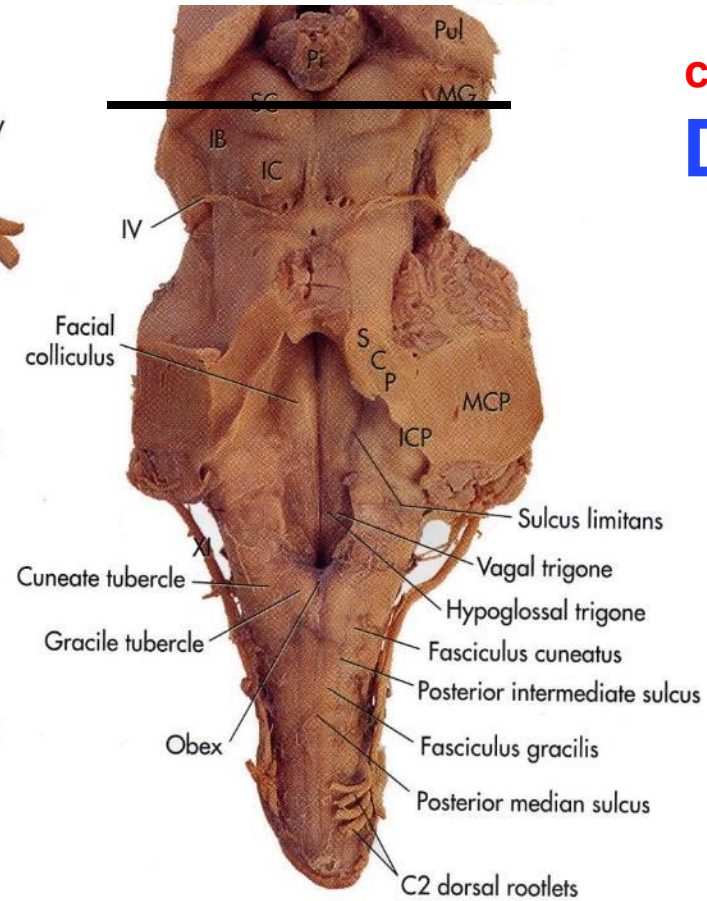
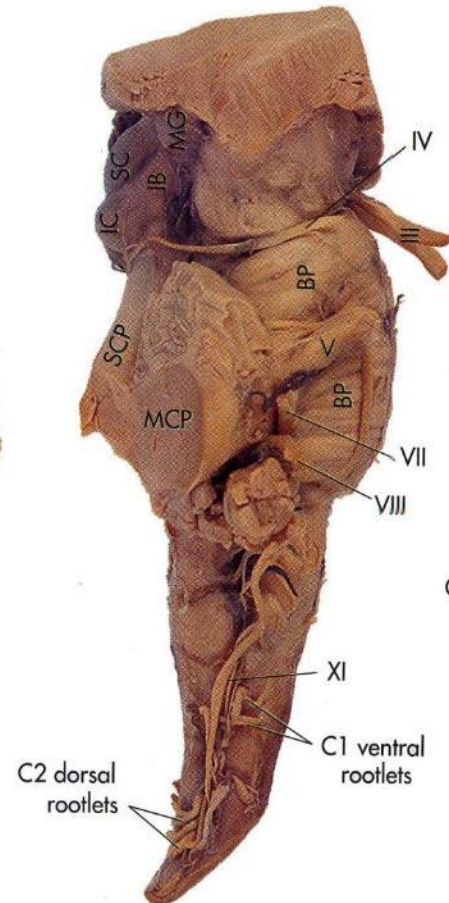
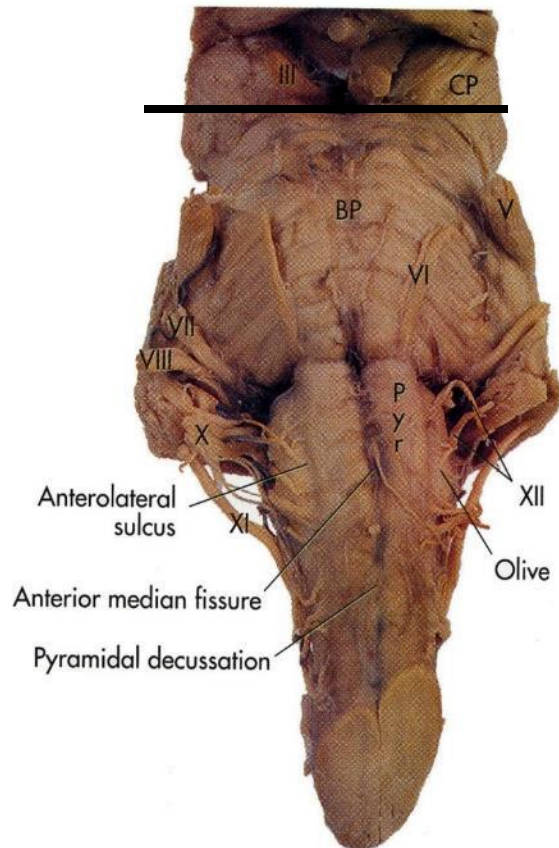




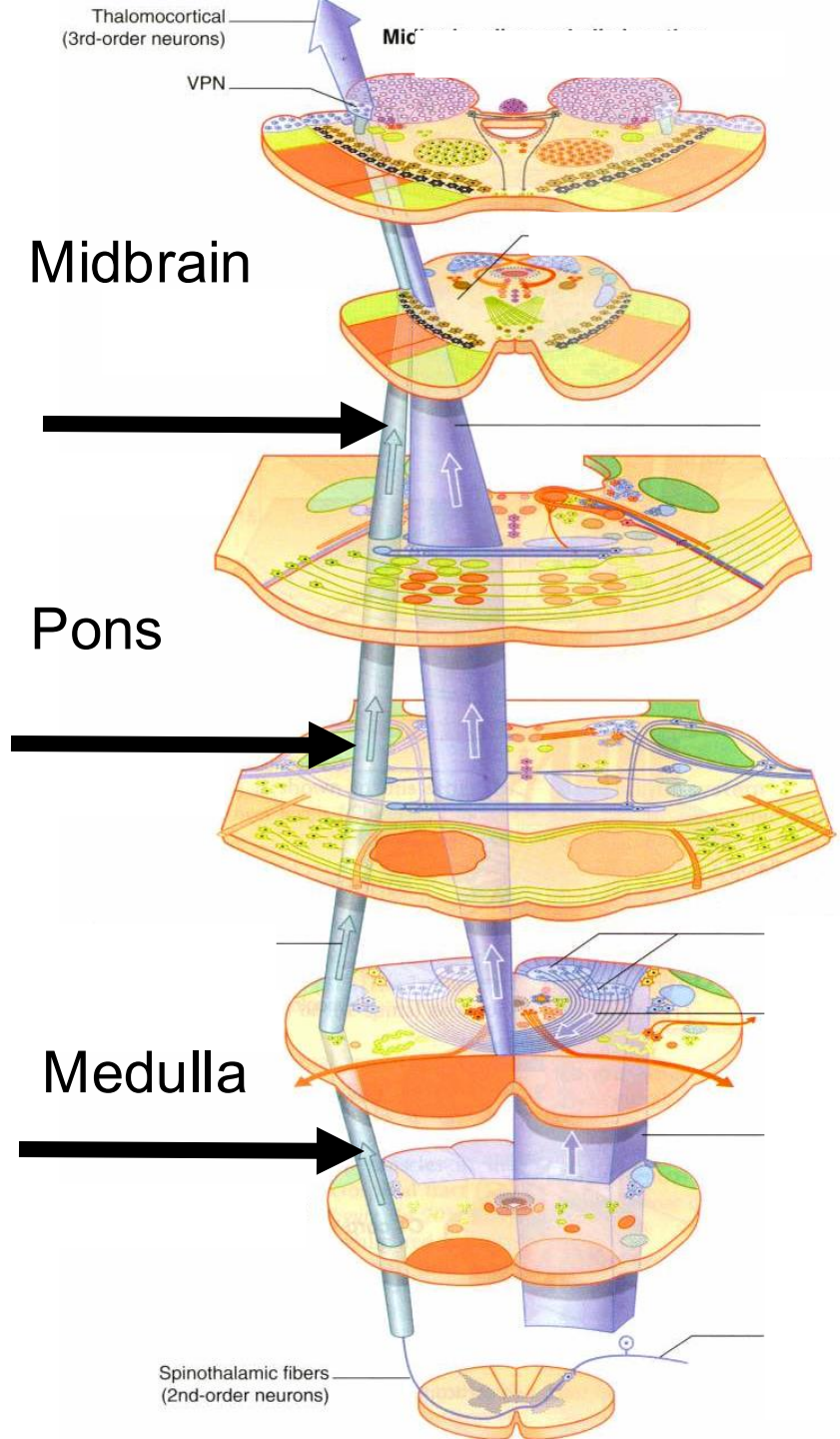
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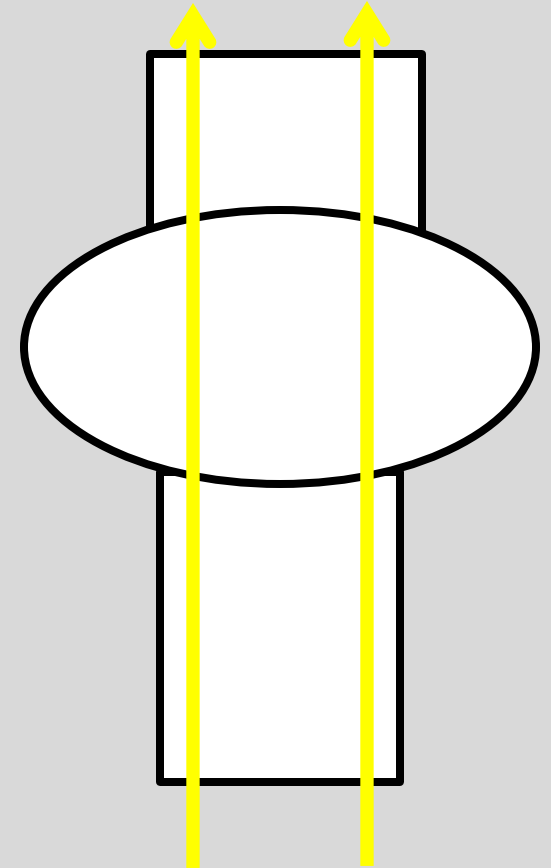


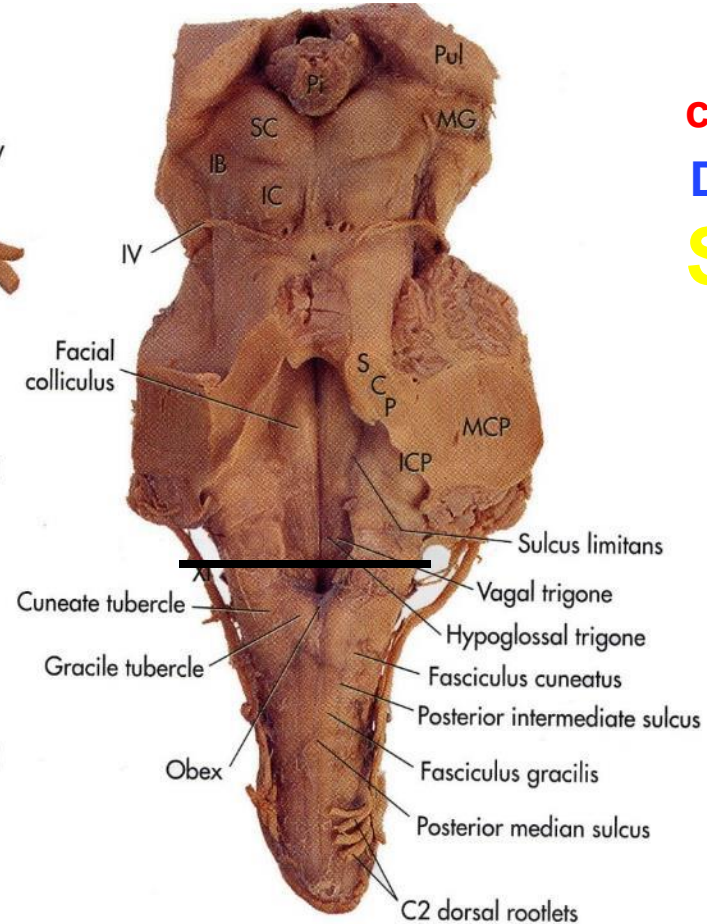
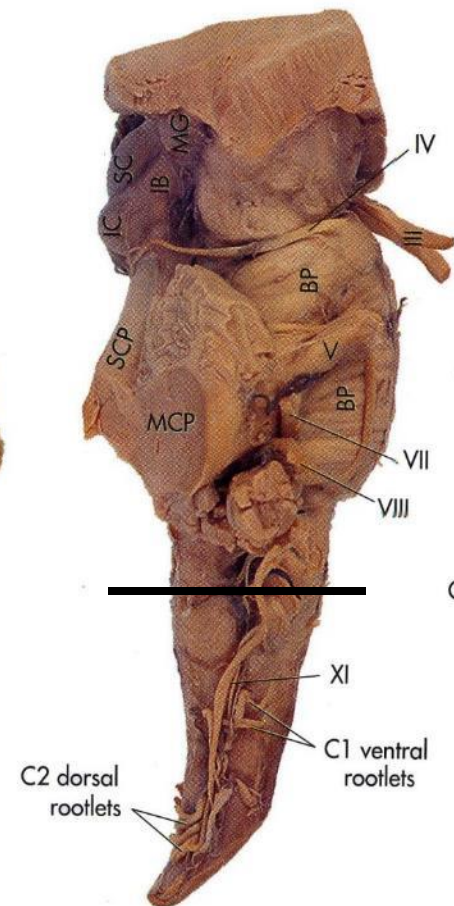
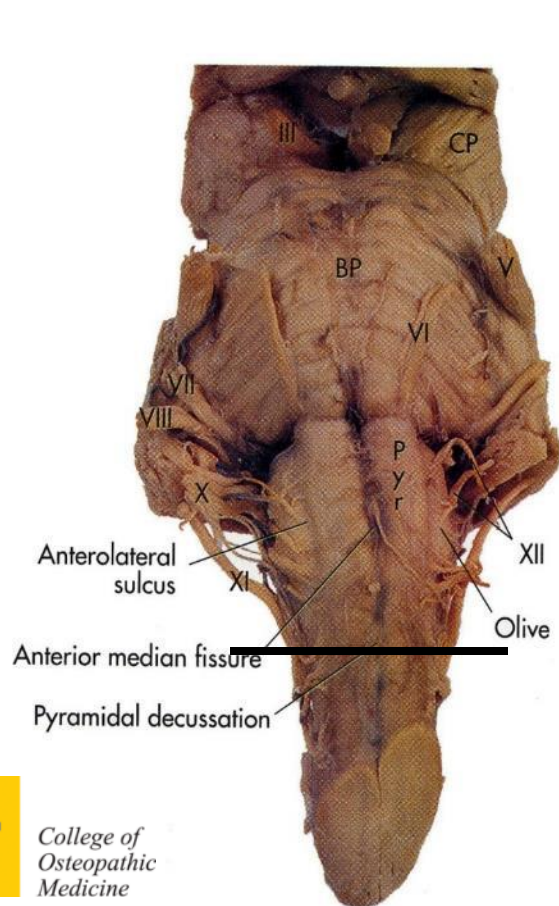
Spinothalamic



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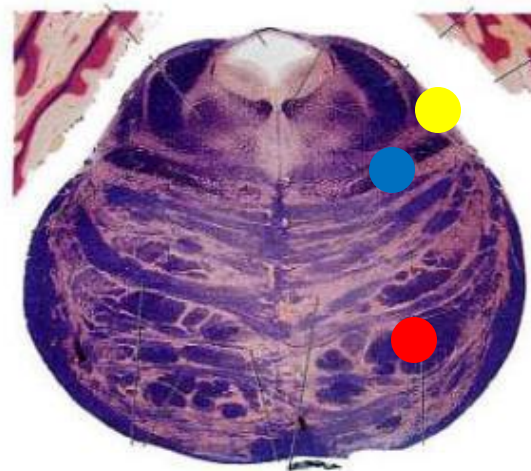




corticospinal

DCML

Spinothalamic



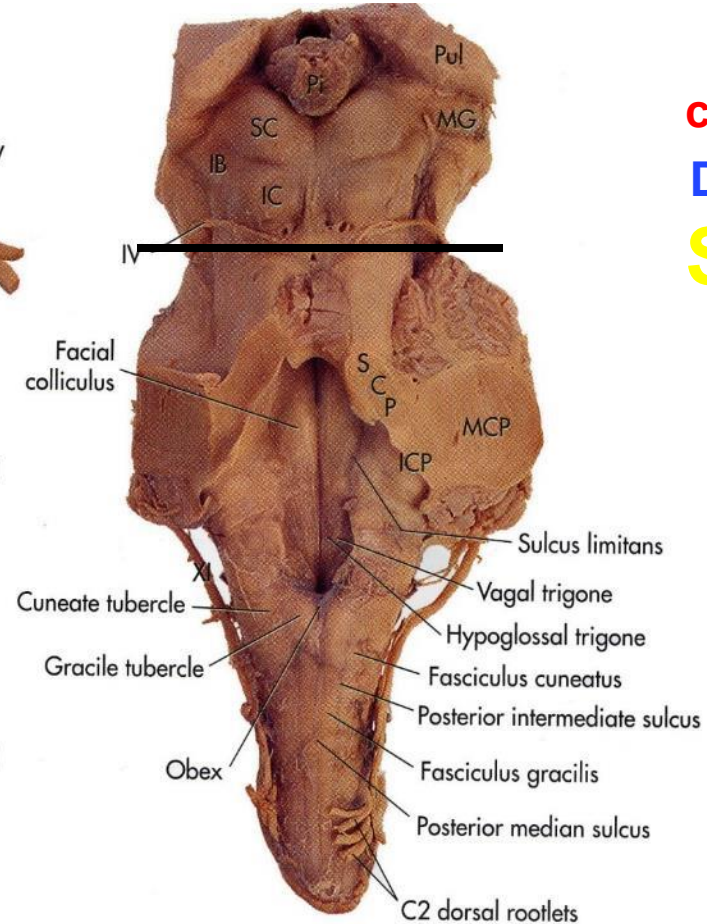
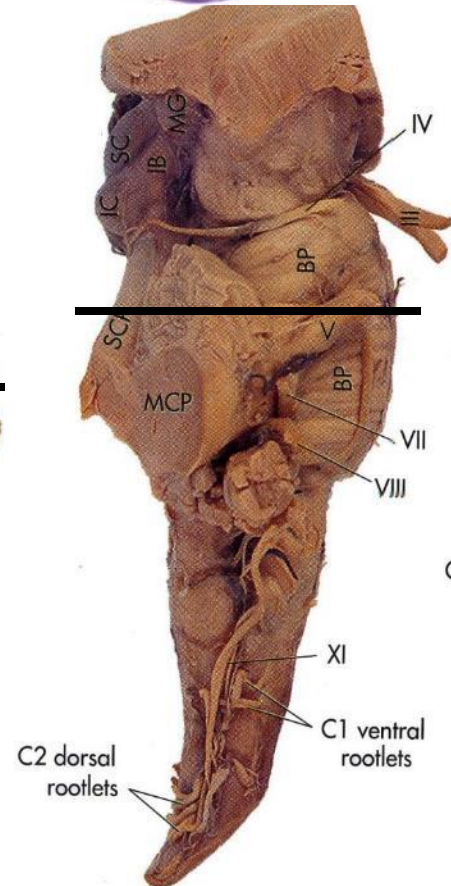
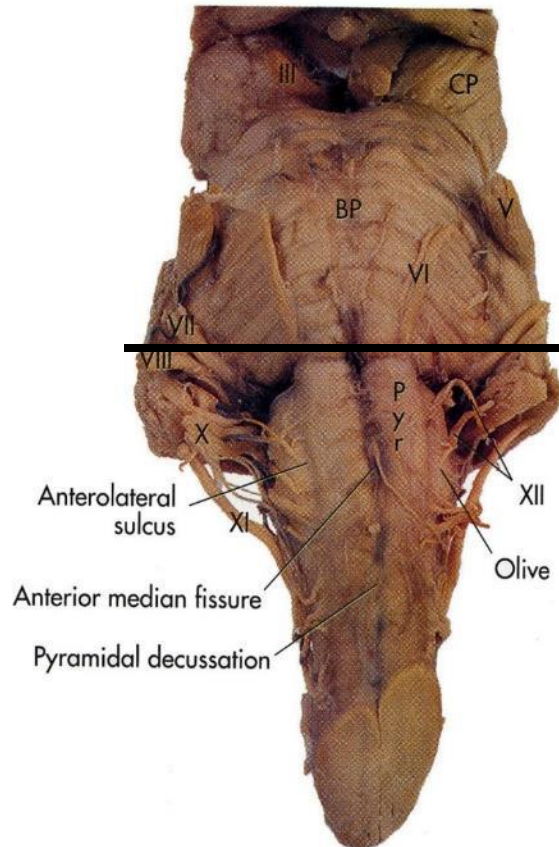
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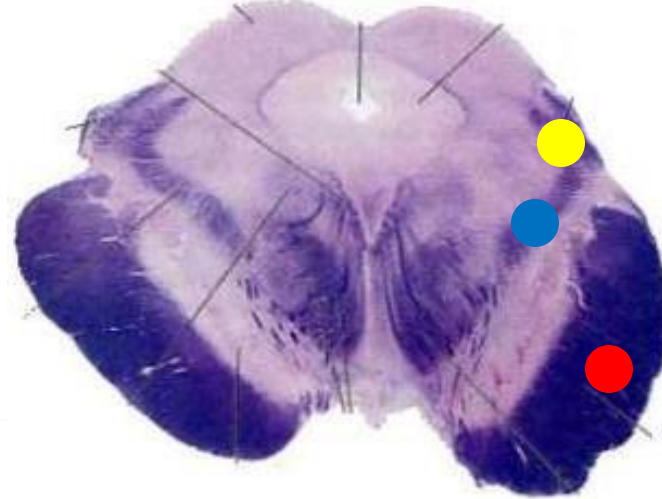
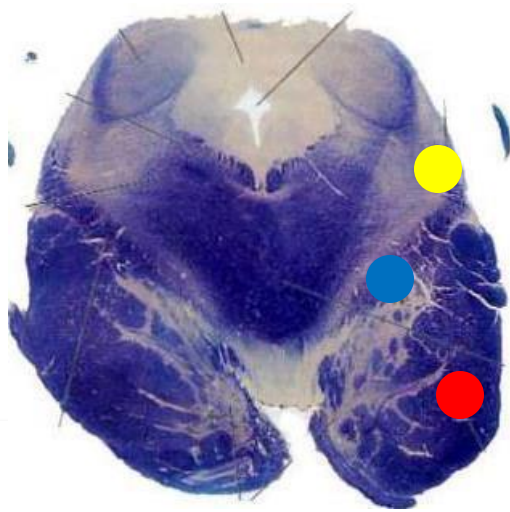
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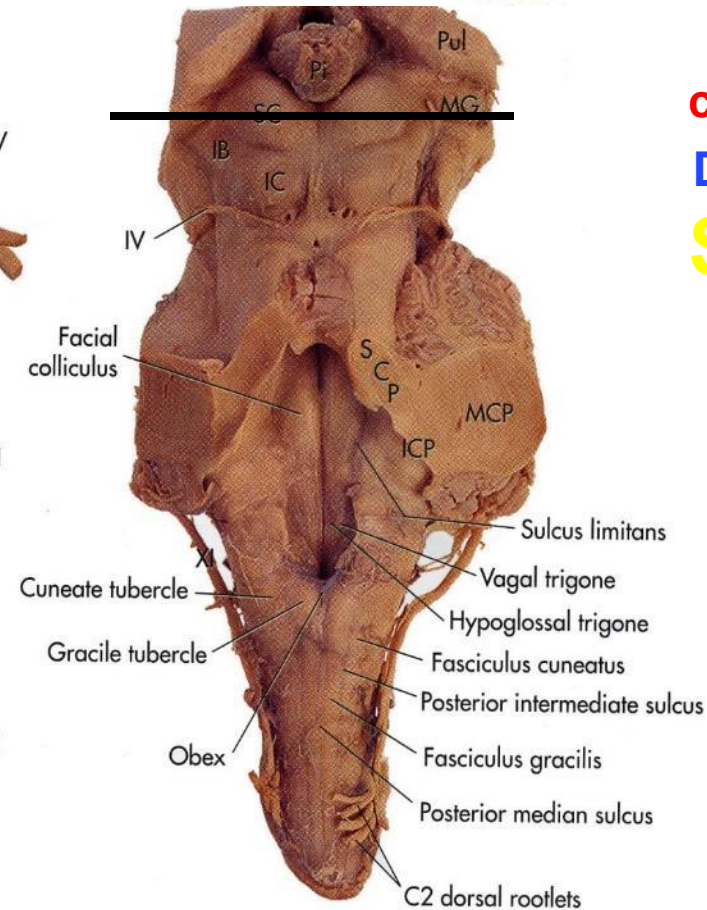
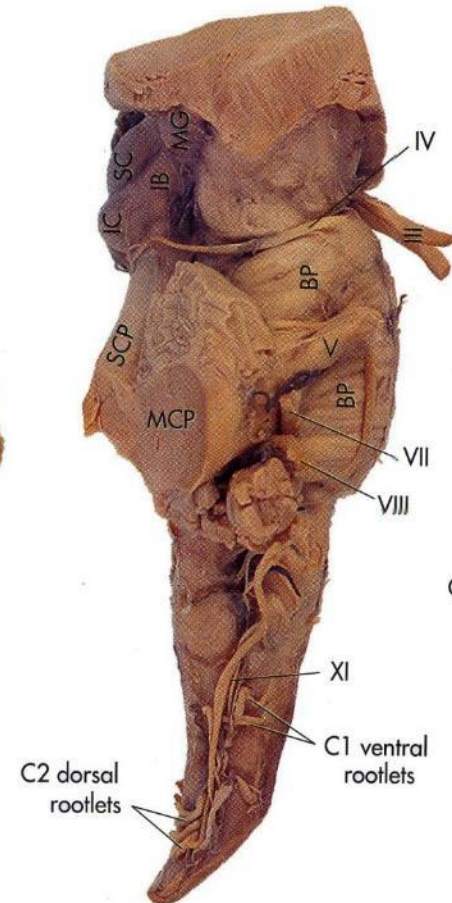
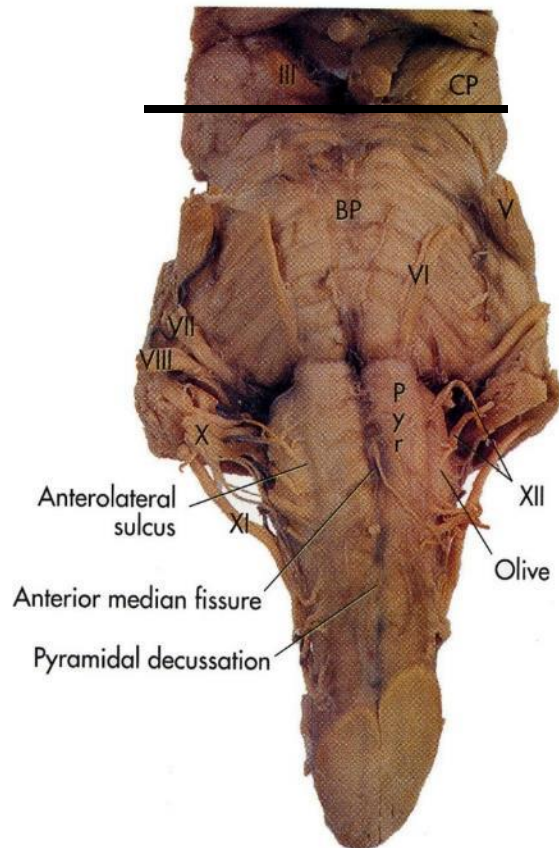
Spinothalamic





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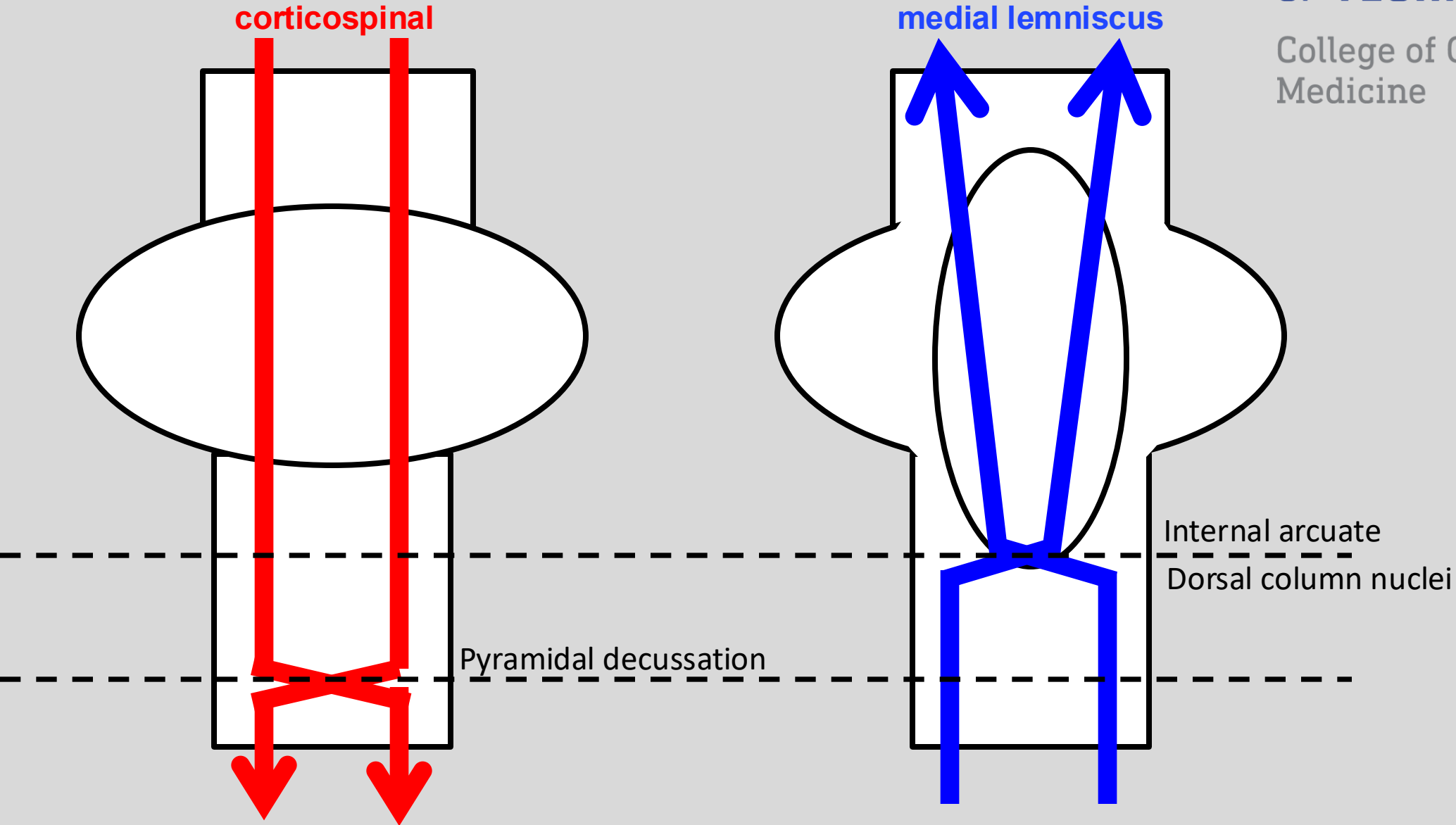
DCML

Spinothalamic

The Corticospinal tract decussates below the DCML decussation

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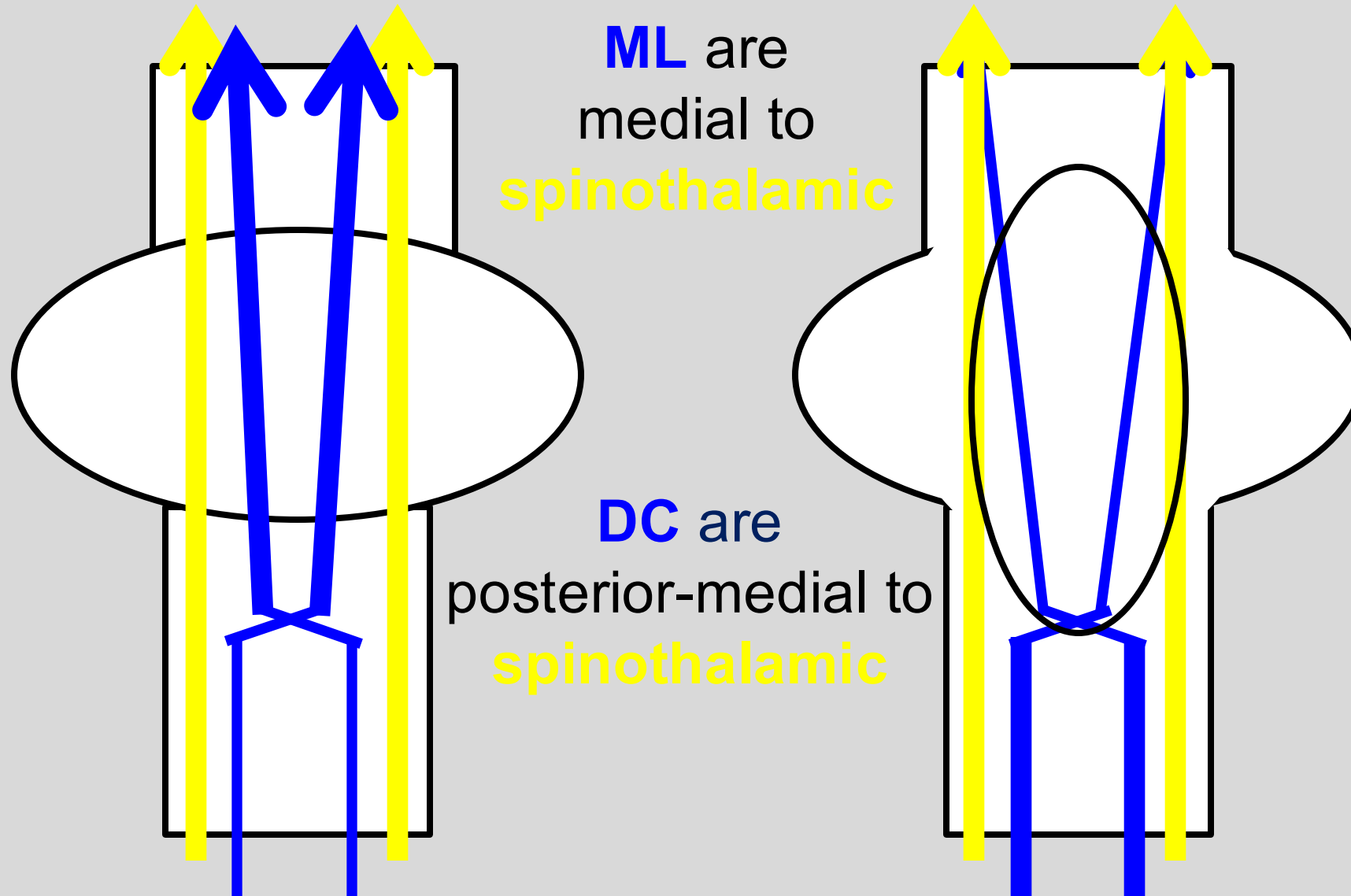
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Relative position of ascending tracts

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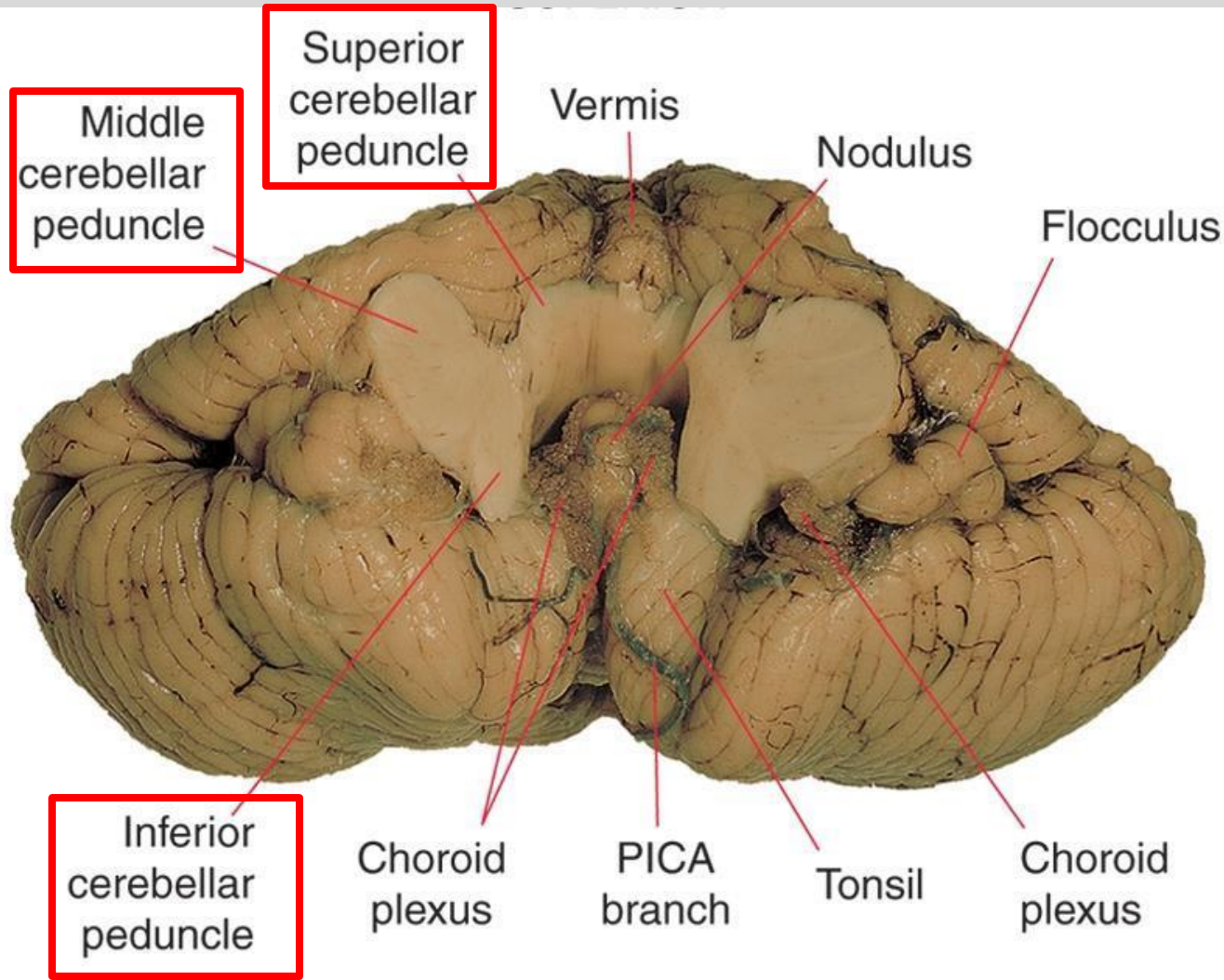
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Cerebellar peduncles

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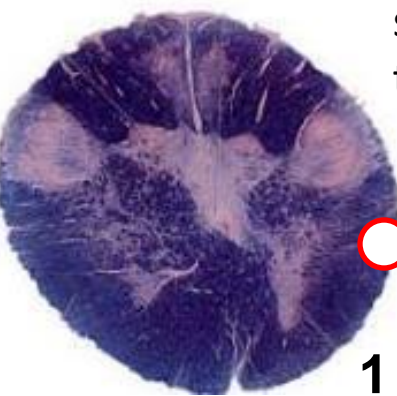
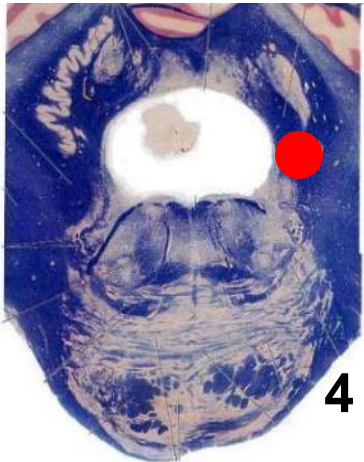
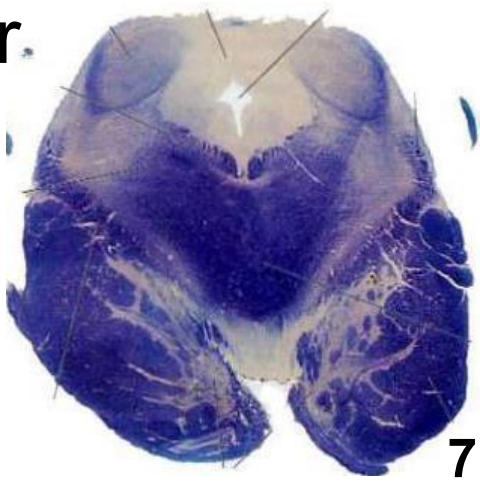
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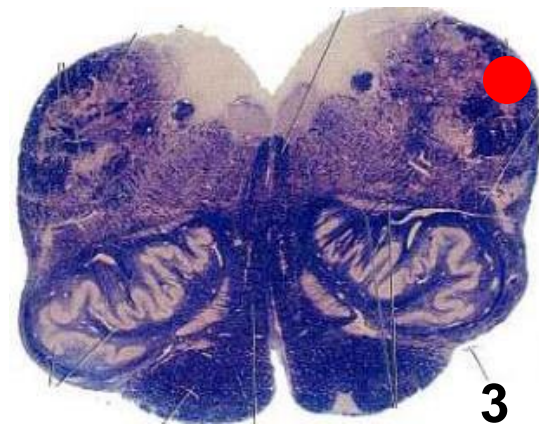
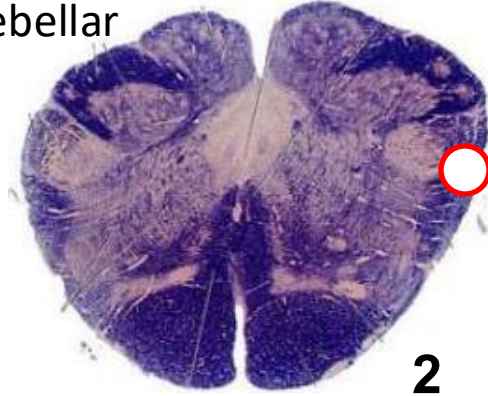
inferior cerebellar peduncle from medulla (afferents)
middle cerebellar peduncle from pons (afferents)
superior cerebellar peduncle to midbrain (efferents)

Inferior cerebellar
peduncle

Inferior

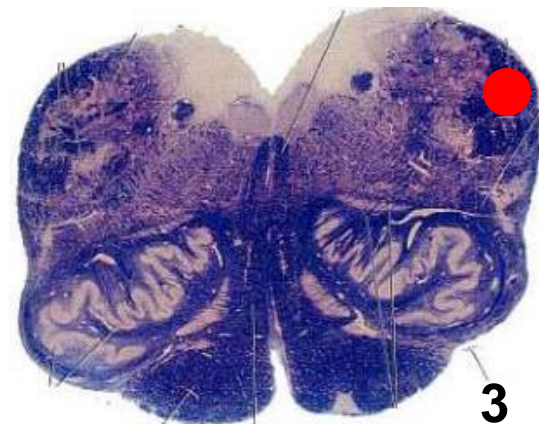
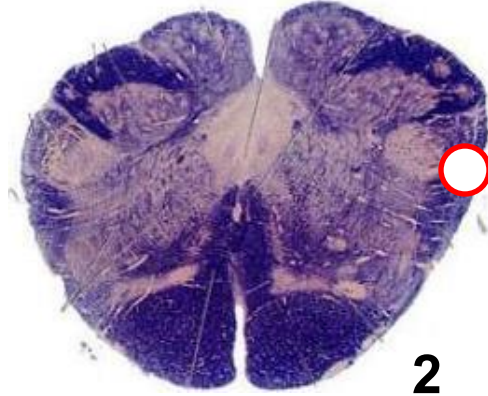
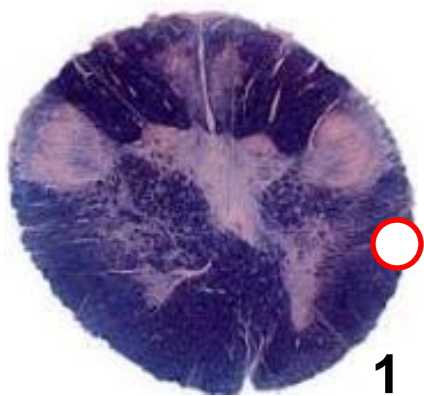
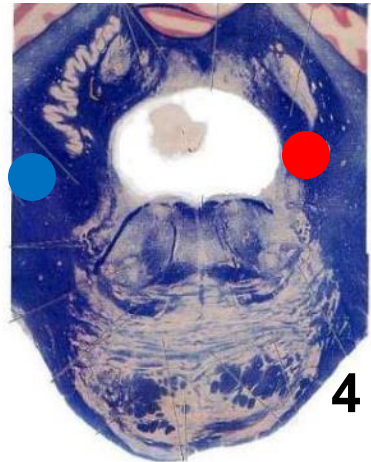
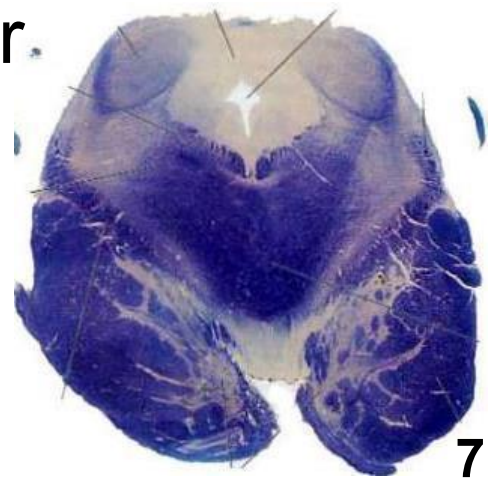


dorsal
spinocerebellar
tract



Middle cerebellar.
Peduncle

Inferior
Middle

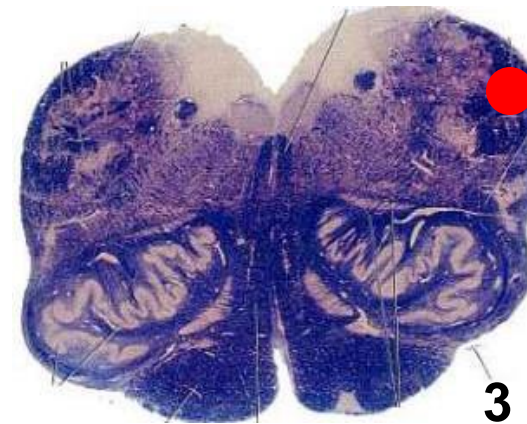
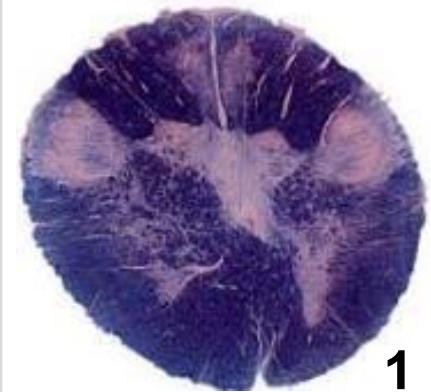
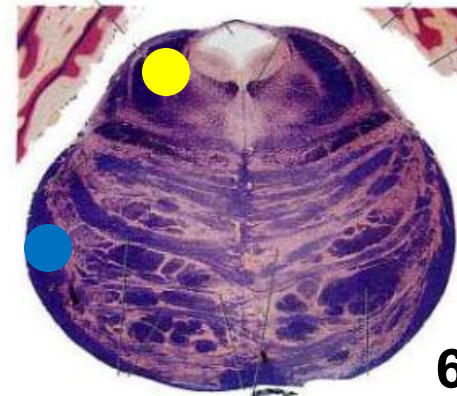
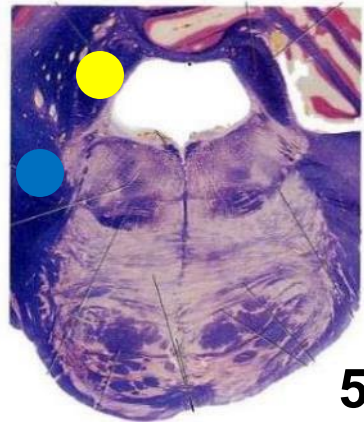
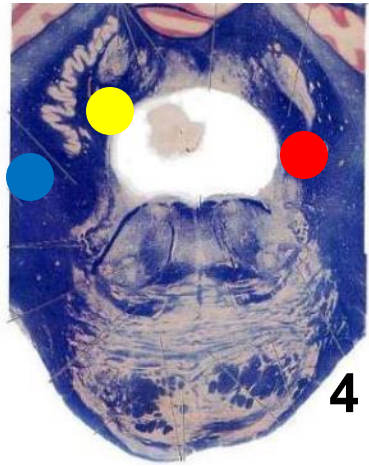
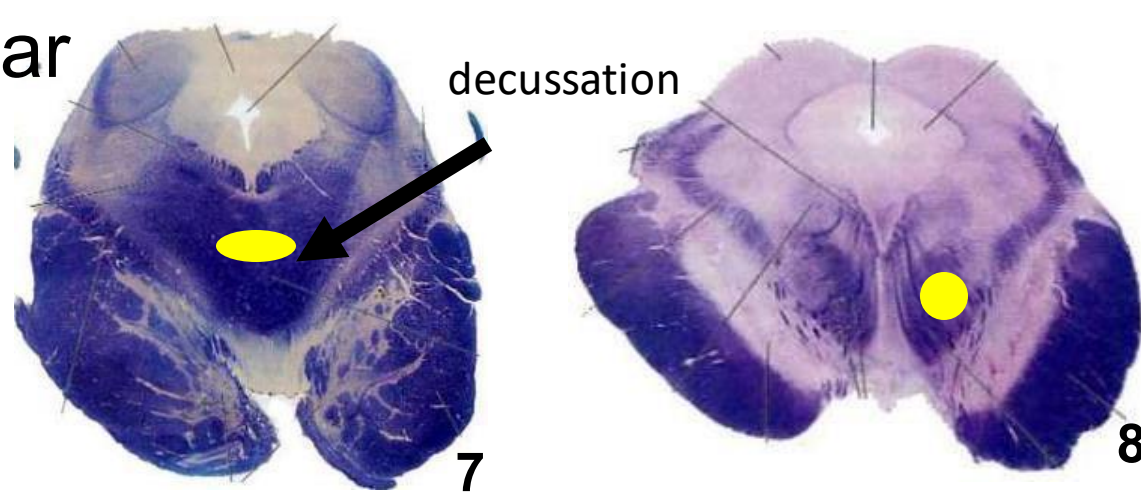


Superior cerebellar peduncle

Inferior

Middle

Superior



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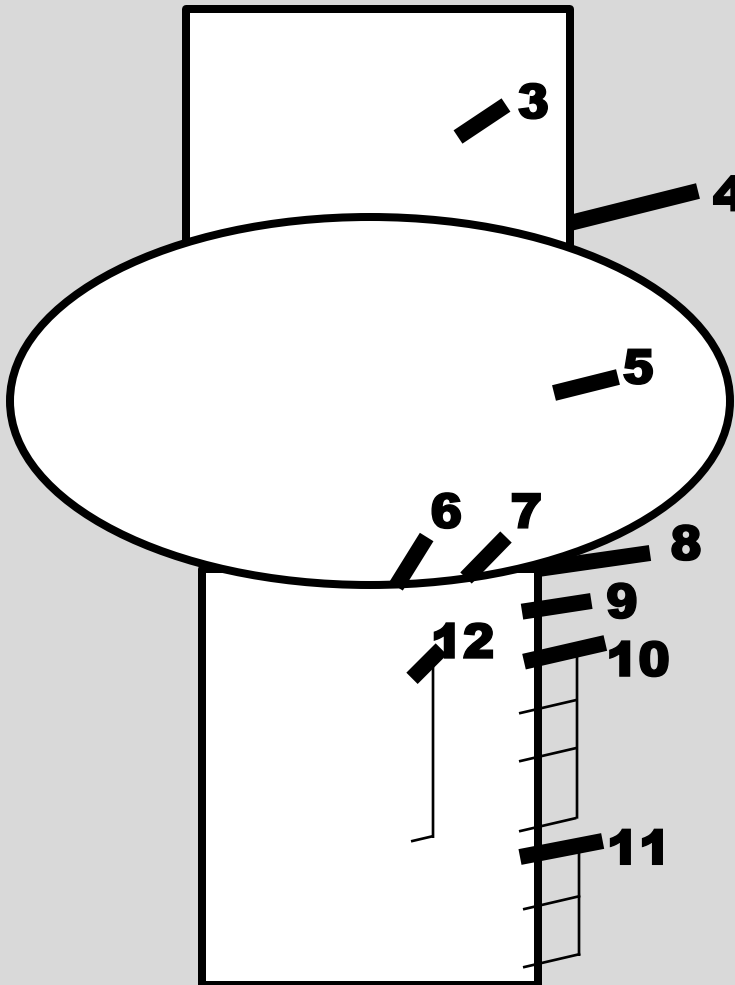
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Cranial nerves

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ventral



3 oculomotor (m)

“Some say

marry

4 trochlear (m)

money,

5 trigeminal (b)

but

6 abducens (m)

my

7 facial (b)

brother

8 vestibulocochlear (s)

says

9 glossopharyngeal (b)

big

10 vagus (b)

brains

11 spinal accessory (m)

matter

12 hypoglossal (m)

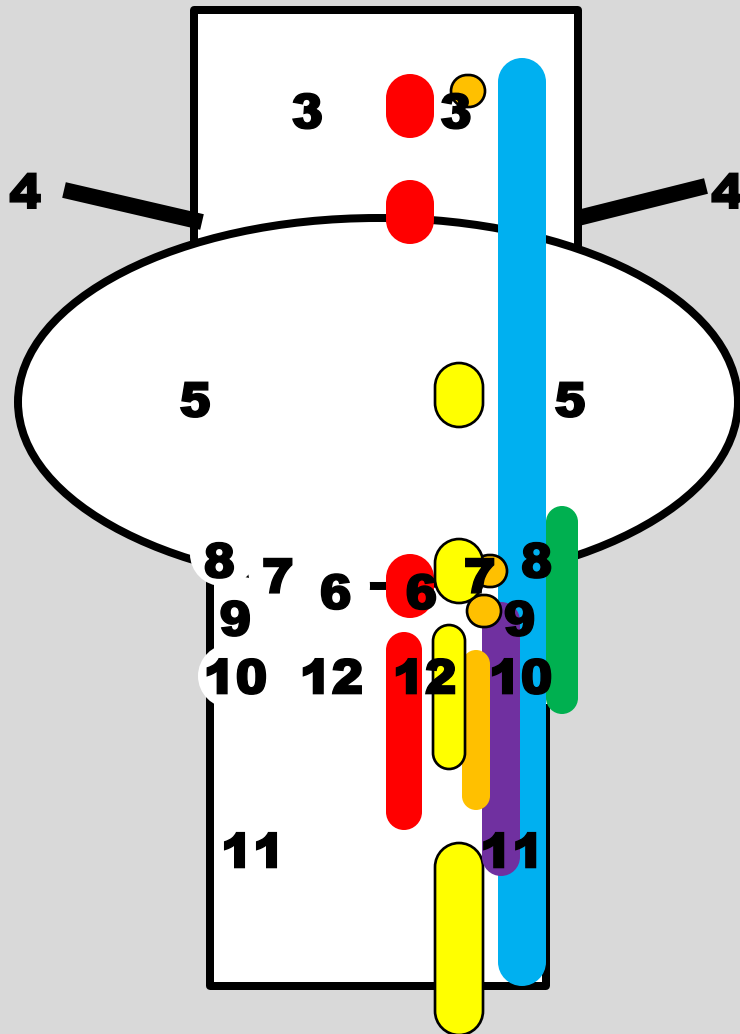
more.”

s = sensory

m = motor

b = both

Cranial nerve nuclei have a consistent rostral-caudal position according to nerve and medial-lateral position according to function



somatic motor

brachial motor

visceral motor

visceral sensory

somatic sensory

special sensory

Motor medial

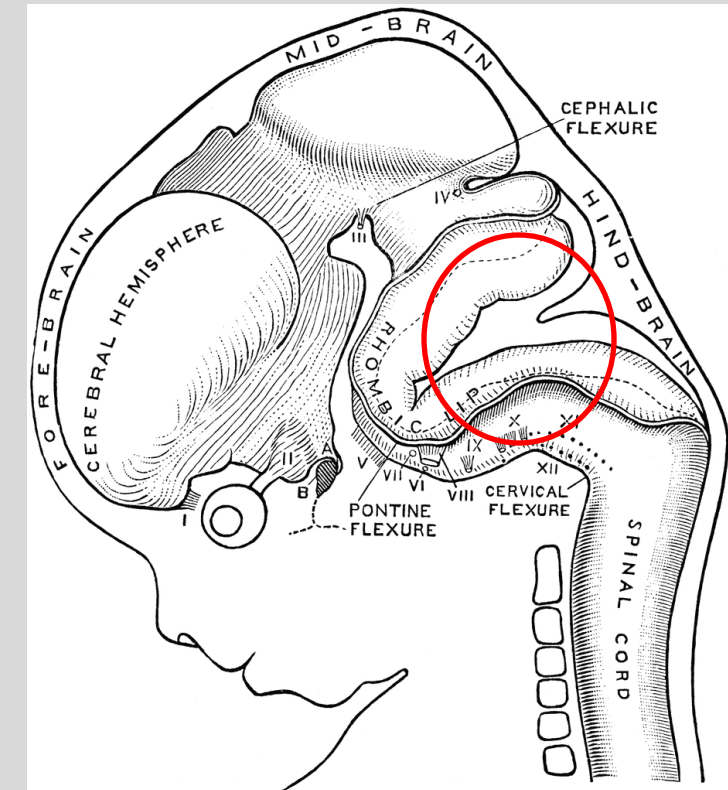
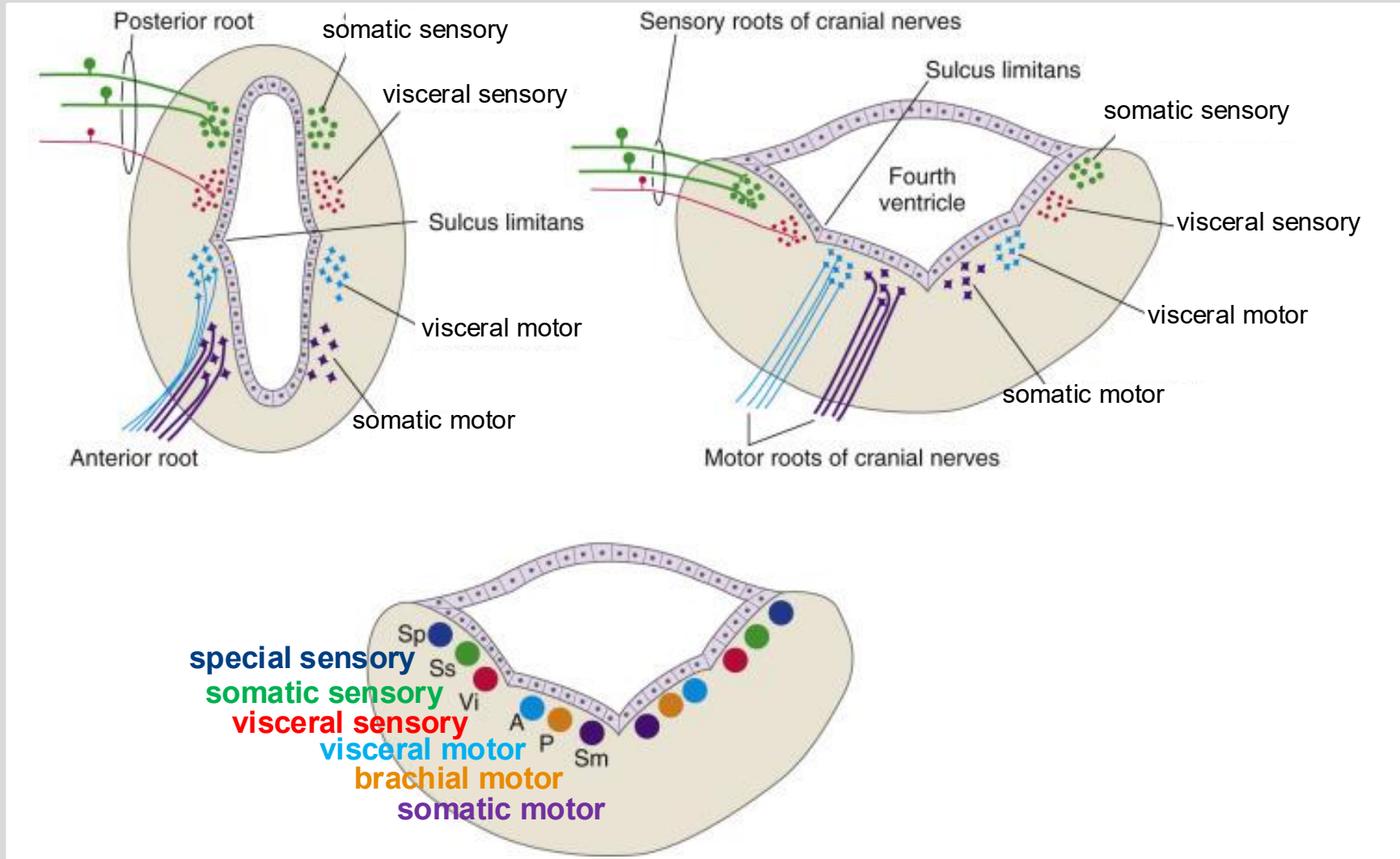
Sensory lateral

- Either motor or sensory, never both (unlike nerves!)
- Partially correspond to the same level as nerves

Developmental basis of nuclei locations

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** Note that colors on this figure do not correspond to other figures*

<i>cranial nerve nuclei summary</i>	<u>SOMATIC MOTOR</u>	<u>BRANCHIAL MOTOR</u>	<u>VISCERAL MOTOR</u>	<u>VISCERAL SENSORY</u>	<u>SOMATIC SENSORY</u>	<u>SPECIAL SENSORY</u>
III	oculomotor → eye muscles		Edinger Westphal → iris			
IV	trochlear → eye muscles					
V		trigeminal motor → jaw			jaw proprio → trigeminal mesencephalic face touch & proprio → trigeminal pontine face pain & temp → trigeminal spinal	
VI	abducens → eye muscles					
VII		facial motor → face & stapedius	superior salivatory → tears & saliva	ant. tongue taste → gustatory	posterior ext. ear → trigeminal spinal	
VIII						orientation → vestibular hearing → cochlear
IX		ambiguus → stylopharyngeus (swallow)	inferior salivatory → parotid gland (saliva)	posterior tongue taste → gustatory carotid baroreceptors → solitary	posterior tongue, tympanic mem → trigeminal spinal	
X		ambiguus → pharynx/larynx (swallow, gag reflex)	dorsal motor vagus → “rest and digest” autonomics	viscerosensory & aortic baroreceptors → solitary	ext. and int. ear → trigeminal spinal	
XI		spinal accessory → scm and upper trapezius				
XII	hypoglossal → tongue					

midbrain	Edinger Westphal	III	pupil constriction	visceral efferent
	oculomotor	III	eyes	somatic motor
	trig. mesencephalic	V	jaw	somatic sensory
	trochlear	IV	eyes	somatic motor
	trigeminal motor	V	jaw, upper neck	branchial motor
	trigeminal pontine	V	face	somatic sensory
	abducens	VI	eyes	somatic motor
	facial motor	VII	face, upper neck	branchial motor
	superior salivary	VII	mouth	visceral efferent
	vestibular	VIII	head motion / orientation	special sensory
pons	cochlear	VIII	hearing	special sensory
	inferior salivary	IX	parotid gland	visceral efferent
	solitary gustatory	{ VII	taste	visceral sensory
		{ IX, X	taste	visceral sensory
	solitary	{ IX	blood pressure, [O2]	sensory
		{ X	respiratory, cardio, digestive	visceral sensory
	hypoglossal	XII	tongue	somatic motor
	ambiguus*	{ IX, XI	swallowing	branchial motor
		{ X	throat, vocals	branchial motor
	spinal accessory**	XI	neck, shoulders	branchial motor
medulla	dorsal vagus	X	gastrointestinal, heart	visceral efferent
		{ V	face, oral cavity	somatic sensory
	trigeminal spinal	{ VII	ear	somatic sensory
		{ IX	throat, vocals	somatic sensory
		{ X	ear, pharynx	somatic sensory

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* ambiguus has a brachial component and a visceral efferent component to inhibit heart rate

**spinal accessory is sometimes classified as brachial and sometimes as somatic motor.

Summary

- The major descending and ascending tracts have distinct trajectories through the brainstem
- The corticospinal tract is always most ventral and decussates in the lower medulla
- The DCML decussates in the upper medulla and changes position through the brainstem but always more medial to the spinothalamic tract.
- The spinothalamic tract decussates in the spinal cord and is always lateral to the DCML.
- The three cerebellar peduncles are present at different levels of the brainstem; inferior – medulla and lower pons; middle – pons; superior – upper pons and lower midbrain.
- The cranial nerve nuclei have a rostral-caudal placement roughly aligned with the cranial nerves.
- Rational arrangement of cranial nerve nuclei: three motor columns are medial and three sensory columns are lateral.

Feedback is appreciated!

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

